1,580 AI Authors

What AI researchers said in December 2024: uncertain, unsafe, and sooner than we thought

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One survey, four rounds

The same questions, put to AI researchers again and again.

1,580

AI researchers

who published in 2023
at NeurIPS, ICML, ICLR,
AAAI, JMLR or IJCAI



1. Asked in December 2024.

2. Largely the same questions as in 2023.

3. A 10% response rate.

4. To the authors' knowledge, the longest-running series of such surveys.

Grace et al., Advanced AI according to 1,580 researchers: uncertain, unsafe, and sooner than we thought, AI Impacts, September 2026, aiimpacts.org/wp-content/uploads/2026/09/ESPAI2024.pdf.

Four moments make you stop

The story of this survey is four moments. Each one makes you stop.

1. Sooner than we thought.
2. A dot plot that barely moved.
3. A tail that will not go away.
4. What they actually fear.

1

Moment 1

Sooner than we thought

The date for human-level AI keeps moving closer

Human-level means every task, and cheaper

The definition every respondent was given.

> *"When unaided machines can accomplish every task better and more cheaply than human workers."*

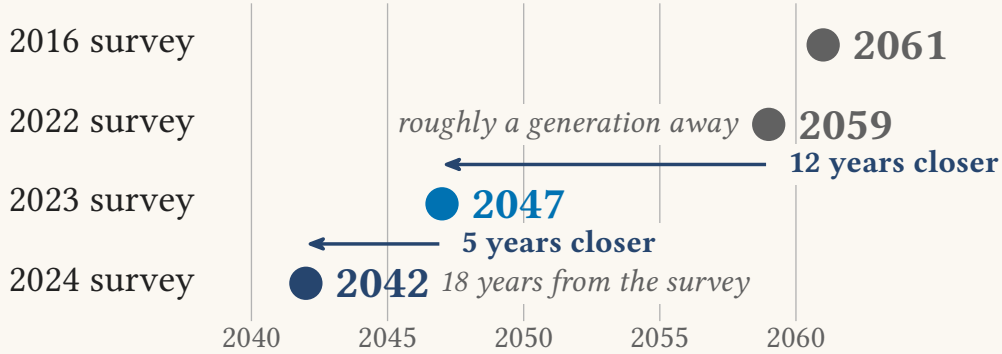
Think feasibility, not adoption.

1. They call it high-level machine intelligence, HLMI.

2. Call it AGI if you prefer.

The 50% date for HLMI keeps moving closer

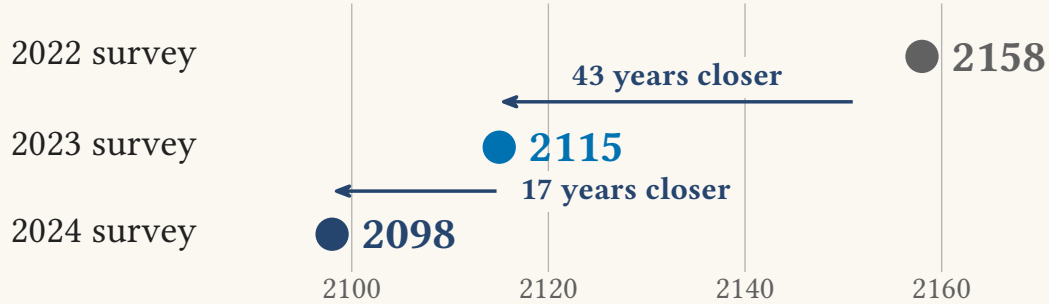
Each survey puts the 50% date of human-level AI closer.



The 2022 date was reported as 2060 in the 2023 paper; the 2024 report recomputes it as 2059.

For every occupation the date moved too

FAOL, full automation of labor: all occupations fully automatable.



Read that again: the 50% date for automating every occupation moved from 2115 to 2098.

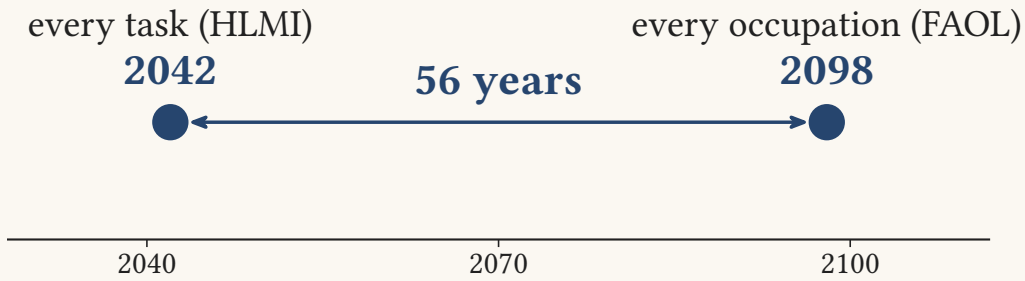
Four dates, a year apart

Year at which the mean aggregate forecast reaches the stated probability.

	2023 survey	2024 survey	Shift
HLMI, 10% probability	2027	2027	0 years
HLMI, 50% probability	2047	**2042**	**5 years**
FAOL, 10% probability	2037	2035	2 years
FAOL, 50% probability	2115	**2098**	**17 years**

The gap to HLMI is 56 years, the smallest yet

The 50% dates for every task and for every occupation, in 2024.



In every survey so far, the two dates are more than fifty years apart.

2

Moment 2

A dot plot that barely moved

Thirty-nine tasks, and forecasts that held

HLMI is too abstract

So they asked about 39 specific things.

1. Write a *New York Times* best-seller novel.
2. Fold laundry.
3. Translate a newly discovered language.
4. Beat the best StarCraft II players.
5. Install the electrical wiring in a new home.

1. Each respondent saw a random four of them.

2. For each: when does the probability cross 50%?

Thirty-nine dots, one per task

Each dot is a task, placed at the year its forecast reaches 50%.



The earliest four are forecast for 2026

The four tasks with the earliest 50% dates.

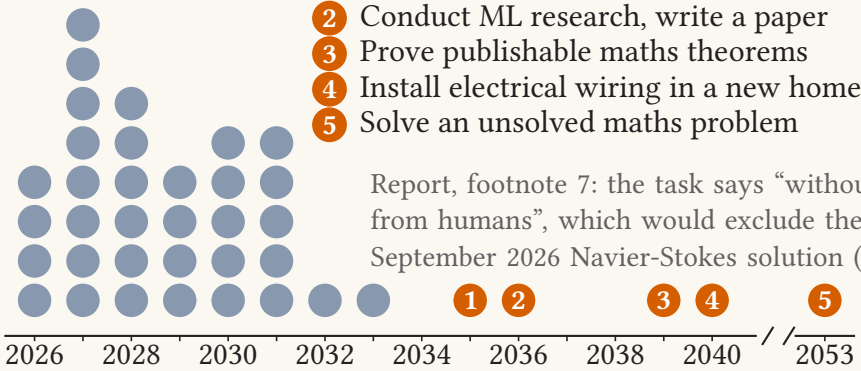


1. Write a high-school history essay — **2026**
2. Write Python code (e.g. quicksort) — **2026**
3. Play new Angry Birds levels (superhuman) — **2026**
4. Win World Series of Poker — **2026**

The report: for tasks with very short timelines "it is plausible that the task is already feasible".

34 of the 39 are at even odds by 2034

Ten years from the survey is 2034. Five tasks come later.



1. Discover physics from simulation — **2035**
2. Conduct ML research, write a paper — **2036**
3. Prove publishable maths theorems — **2039**
4. Install electrical wiring in a new home — **2040**
5. Solve an unsolved maths problem — **2053**

Report, footnote 7: the task says “without more input from humans”, which would exclude the proposed September 2026 Navier-Stokes solution (OpenAI, 2026).

33 of the 39 moved less than two years

Timelines were already short, and changed little.

33
of the 39 forecasts

1. Each moved by less than two years, in either direction.

2. The two largest moves, both roughly seven years earlier: proving publishable maths theorems and conducting ML research.

3

Moment 3

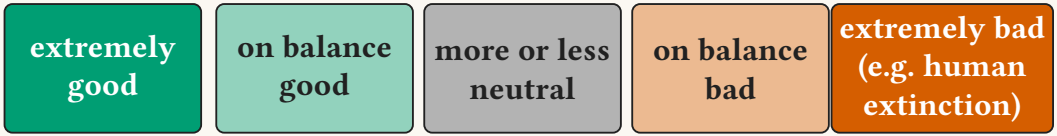
A tail that will not go away

How likely they think the worst outcomes are

They were also asked about outcomes

The exact question, asked of all respondents (n = 1,538).

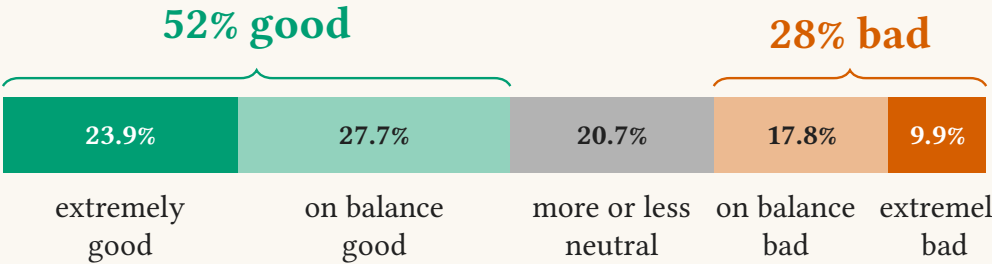
> *"Assume for the purpose of this question that HLMI will at some point exist. How positive or negative do you expect the overall impact of this to be on humanity, in the long run?"*



Respondents split 100% across the five kinds of impact.

On average, 52% good and 28% bad

The five kinds of impact, averaged over all respondents.



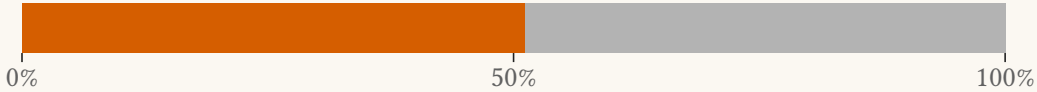
1. The median respondent puts 5% on extremely bad.

2. 788 of 1,538 respondents put at least 5% on both extremes.

The number that stops a boardroom

Human extinction or similarly permanent and severe disempowerment.

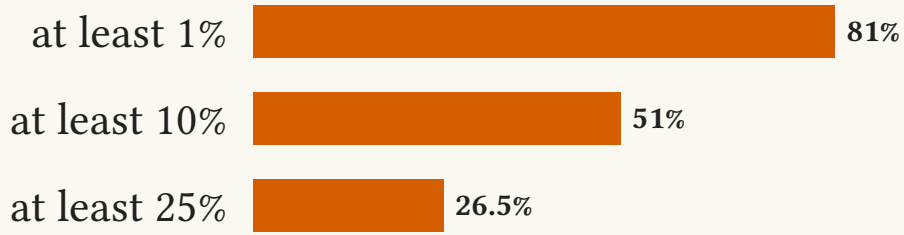
51% of surveyed AI researchers give it at least **10%**.



Pooled over three question variants (n = 1,489): 51.1%.
Each variant alone: 49.0% to 52.7%.

81% give it at least 1%, 51% at least 10%

Share of researchers by the chance they give it.

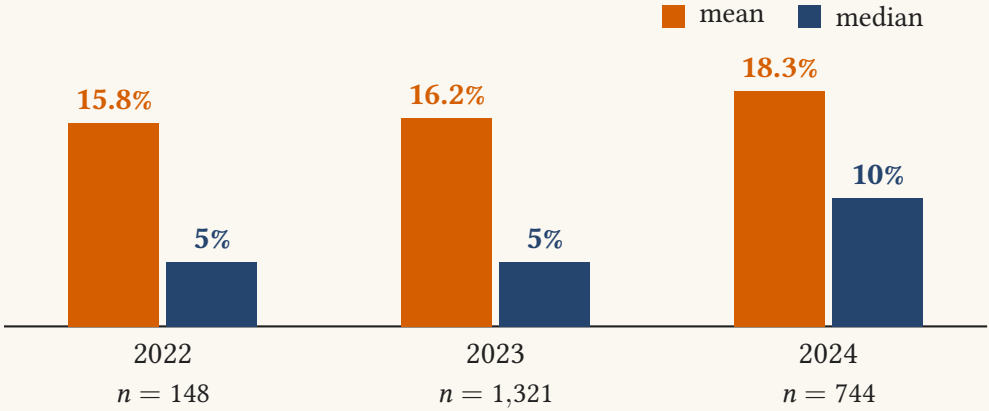


1. 12% gave it zero chance. A third gave it at least 20%.

2. The middle half of the estimates ranged from 1% to 25%.

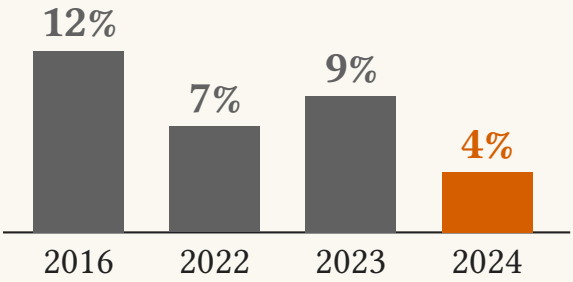
For the first time the median is 10%

Chance of extinction or similar, no further conditions.



Intelligence explosion: 4% say quite likely

Share who find the argument quite likely, 81 to 100%.



The argument: if AI does nearly all research and development, progress could become more than an order of magnitude faster in under five years.

4

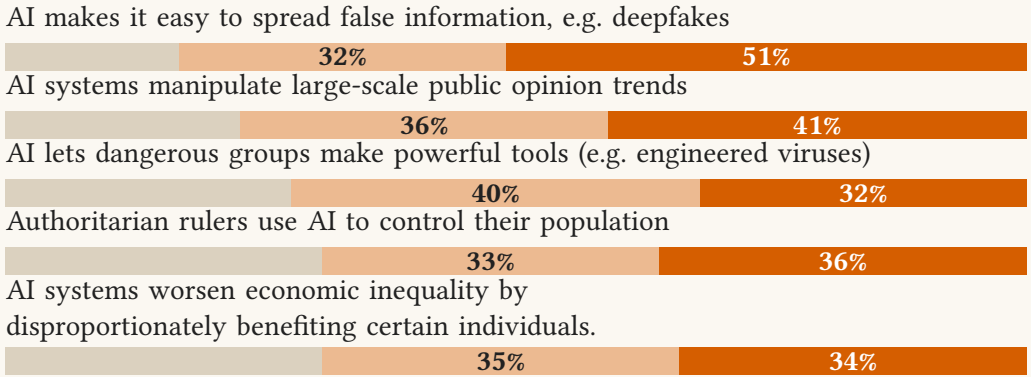
Moment 4

What they actually fear

What worries them, and what they want done

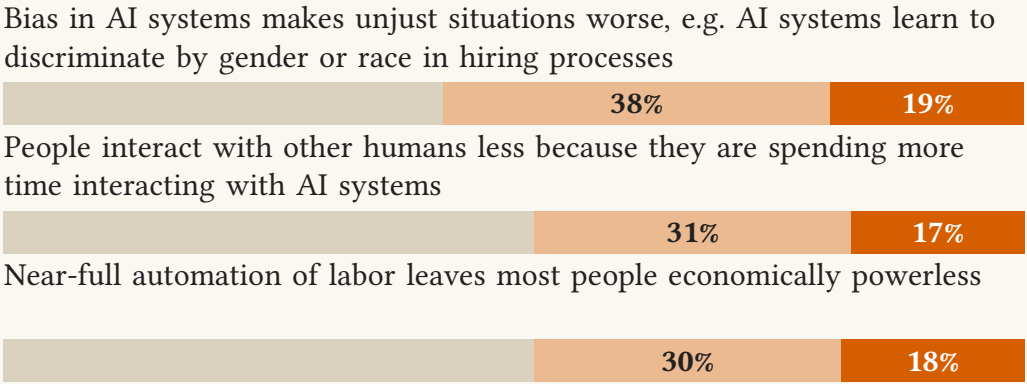
Two scenarios top the list of concerns

Concern over the next thirty years. **deepfakes: 83% substantial or extreme**



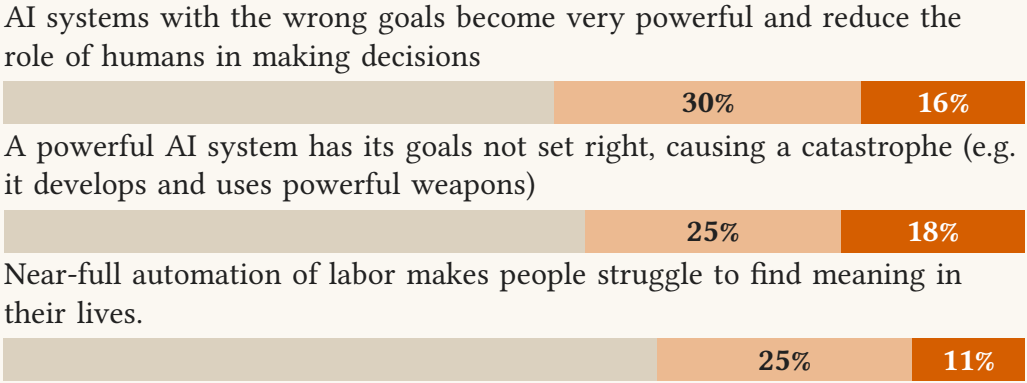
Three more, still of substantial concern

Concern over the next thirty years, continued.



The last three, lost meaning at the bottom

The least, lost meaning, still worries over a third of them.



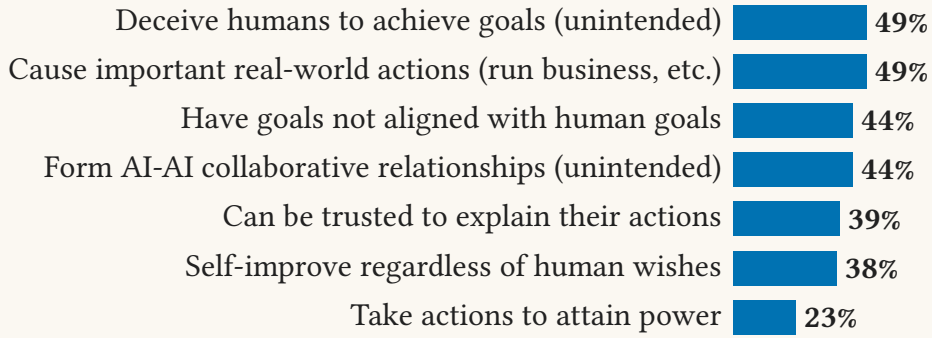
Four traits that most expect by 2044

Share who rate each trait likely or very likely for some state-of-the-art AI.



Seven traits that fewer expect, power the least

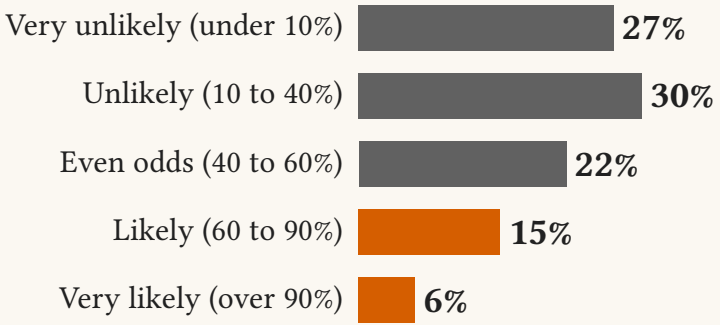
The same question: likely or very likely in 2044.



Up since 2023 for almost every trait.

Only 22% expect to know why an AI decided

Typical state-of-the-art systems in 2029 (n = 520).

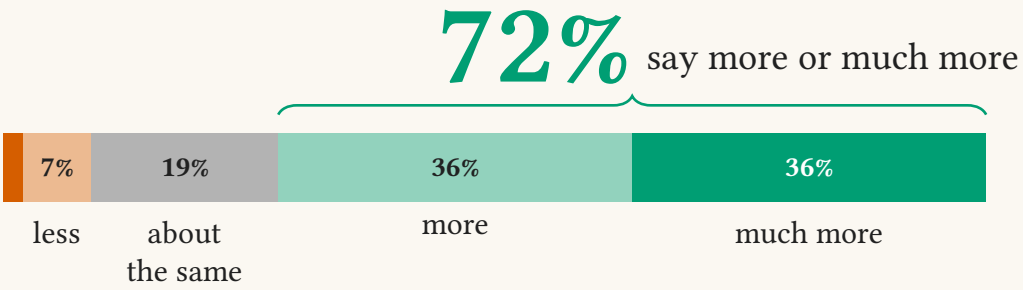


22%
likely or very likely

The bars are rounded; the report's text gives 22% for the two together.

And what they want spent on it

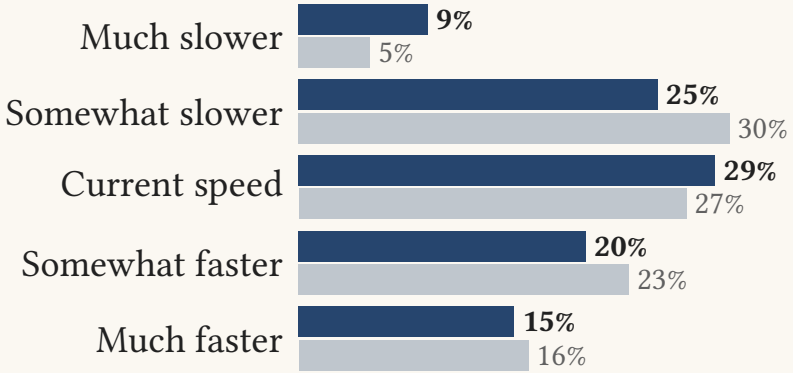
Relative to how much society prioritises AI safety research now (n = 748).



The thin sliver at the left is "much less", which the figure leaves unlabelled.

A third each: faster, slower, or as now

The pace that would make them most optimistic for humanity's future.



5

Closing

So what?

What to take from the survey

Four limits the report states itself

The authors' own words on what this survey can tell.

1. **A 10% response rate.**
2. **AI-company researchers are almost certainly underweighted.**
   The authors: timelines may have shrunk more than shown.
3. **"Expertise, however, is a poor predictor of forecasting accuracy."**
4. **The survey is from December 2024.**
   The report: it "was conducted in December 2024, over 18 months ago".

Three things to leave this room with

What the survey says, and what to do with it.

1. **Expert timelines keep moving.**
   The HLMI 50% date: 2061, 2059, 2047, 2042. Do not anchor a five-year plan on a point estimate.

2. **Narrow-task forecasts were already near.**
   34 of 39 tasks at even odds within ten years. Some early dots may already be behind us, as the report itself notes.

3. **The tail is not a fringe.**
   51% of the field assigns at least 10% to extinction or similarly permanent and severe disempowerment.

The one number: 3.4

From 2016 to 2024, the 50% date for human-level AI.

3.4

years closer,
for each year that passed.

The HLMI date went from *45 years out to 18* in eight years.

Plan as if the date could keep moving.

Every quoted phrase, with page number (1)

- **HLMI definition:** “unaided machines can accomplish every task better and more cheaply than human workers.” (first letter capitalised on the slide) and “Think feasibility, not adoption.” Report p. 5; the question is on p. 12.
- **FAOL definition:** “when all occupations are fully automatable.” Report p. 5.
- **Outcomes question:** “Assume for the purpose of this question that HLMI will at some point exist. How positive or negative do you expect the overall impact of this to be on humanity, in the long run?” Report p. 19, asked of all respondents (n=1538).
- **Five kinds of impact:** “Extremely good (e.g. rapid growth in human flourishing) … On balance good … More or less neutral … On balance bad … Extremely bad (e.g. human extinction)” Report p. 21.

Every quoted phrase, with page number (2)

- **The 51%:** “51% of researchers assign at least a 10% chance to human extinction or similarly permanent and severe disempowerment from advanced AI” Report p. 3 (Figure 1 caption); pooled 51.1% in Table 4, p. 27.
- **Very short timelines:** “it is plausible that the task is already feasible” Report p. 10.
- **Footnote 7:** “without more input from humans” and “the proposed September 2026 Navier-Stokes solution” Report p. 9, footnote 7.
- **Two scenarios:** “AI makes it easy to spread false information, e.g. deepfakes” and “AI systems manipulate large-scale public opinion trends” Report p. 19; the 83% is in the abstract, p. 1; Figure 12, p. 21.

Every quoted phrase, with page number (3)

- **The limits:** “Expertise, however, is a poor predictor of forecasting accuracy.” and “was conducted in December 2024, over 18 months ago” Report p. 34; the sentence on company researchers is on the same page.
- **The one number:** “getting about 3.4 years closer for each year that passed” Report p. 1.

Where each date comes from

- **HLMI, 50% date:** 2061 (2016), 2059 (2022), 2047 (2023), 2042 (2024). Report p. 1, footnote 1: 2022 was reported as 2060 in the 2023 paper. The drops of twelve and five years are on p. 12; Table 1 is on p. 15.
- **FAOL, 50% date:** 2158 (2022), 2115 (2023), 2098 (2024). Report p. 12, footnote 8: 2023 was reported as 2116 in the 2023 paper. The drops of 43 and 17 years are on p. 12; the gap of 56 years is on p. 15.
- **The 39 tasks:** the 50% year of each is in Table 7, report p. 45; 34 of 39 by 2034 and the five later ones are on p. 9, 33 of 39 moving less than two years on p. 10.

Where each chart comes from

- **Moment 2, the dots:** one dot per row of Table 7, report p. 45, at its printed 50% year.
- **Moment 3:** Table 3, p. 22; Table 4, p. 27; the text of p. 23; Figure 17, p. 26 (means and medians) and Figure 8, p. 17 (the shares who find the argument quite likely).
- **Moment 4:** the percentages are printed inside Figures 10, 11, 12, 21 and 23 of the report (pp. 20, 21, 30, 31); the text repeats the 83%, 22%, 72%, 34%, 34%, 29%, 5% and 9% (pp. 1, 18, 28, 33).
- **Every chart is redrawn** from these numbers; none of the report's figures is reproduced.

The two papers

- Grace, K., Simonelli, J., Weinstein-Raun, B., Krueger, D., Young, N., Duffy, T., Adamczewski, T., Reeve, J. and Beck, A. (2026). *Advanced AI according to 1,580 researchers: uncertain, unsafe, and sooner than we thought.* AI Impacts, September 2026, 48 pages. The 2024 round of the Expert Survey on Progress in AI. A report on the AI Impacts site; the PDF states no licence.

- Grace, K., Sandkühler, J. F., Stewart, H., Weinstein-Raun, B., Thomas, S., Stein-Perlman, Z., Salvatier, J., Brauner, J. and Korzekwa, R. C. (2025). *Thousands of AI Authors on the Future of AI.* Journal of Artificial Intelligence Research, 84. doi:10.1613/jair.1.19087. The 2023 round, with 2,778 respondents; the earlier deck was built on it.